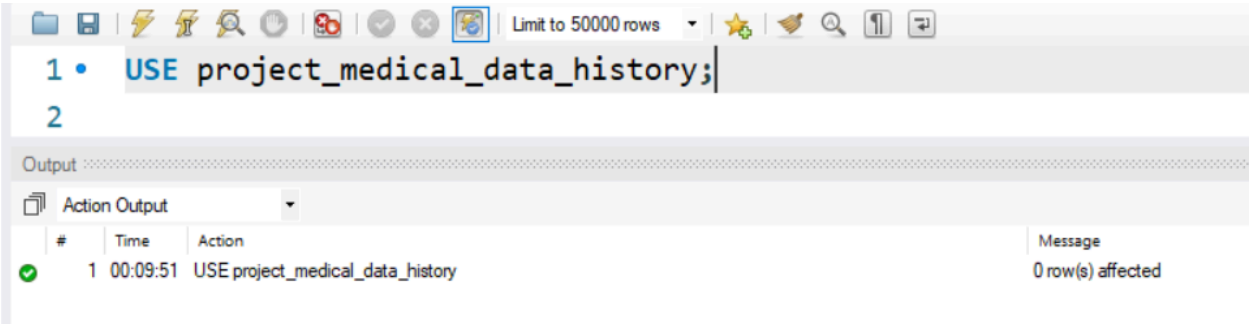
PRSQL-02 Medical Data History

Name: MEHTAB BIBI

Project Team ID: PTID-CDA-MAR-26-1354

Select Database

```
USE project_medical_data_history;
```



See All Tables

```
SHOW TABLES;
```

Output :

Tables_in_project_medical_data_history
admissions
doctors
patients
province_names

Q1. Show first name, last name, and gender of patients whose gender is 'M'

```
SELECT first_name, last_name, gender
```

```
FROM patients
WHERE gender = 'M';
```

This query shows only those patients whose gender is male (M). It selects their first name, last name, and gender from the patients table.

Output :

first_name	last_name	gender
Donald	Waterfield	M
Mickey	Baasha	M
Jiji	Sharma	M
Blair	Diaz	M
Charles	Wolfe	M
Thomas	ONeill	M
Sonny	Beckett	M
Cedric	Coltrane	M
Hank	Spencer	M
Rick	Bennett	M
Woody	Bashir	M
Tom	Halliwell	M
John	West	M
Jon	Doggett	M
Nino	Andrews	M

Q2. Show first name and last name of patients who does not have allergies.

```
SELECT first_name, last_name
FROM patients
WHERE allergies IS NULL;
```

This query finds patients who don't have any allergy recorded. It checks for NULL values in the allergies column and shows their names.

Output :

first_name	last_name
Donald	Waterfield
Blair	Diaz
Thomas	ONeill
Sonny	Beckett
Cedric	Coltrane
Hank	Spencer
Sara	di Marco
Amy	Leela
Rachel	Winterbourne
John	West
Jon	Doggett
Angel	Edwards
Brodie	Beck
Nea	Kobayakawa
Laura	Halliwell

Q3. Show first name of patients that start with the letter 'C'

```
SELECT first_name
FROM patients
WHERE first_name LIKE 'C%';
```

This query selects patients whose first name starts with the letter “C”.

Output :

first_name
Charles
Cedric
Charles
Cross
Calleigh
Catherine
Caroline
Casanova
Chen
Charmian
Cary
Christine
Carol
Corrado
Cordelia

Q4. Show first name and last name of patients that weight within the range of 100 to 120 (inclusive)

```
SELECT first_name, last_name
FROM patients
WHERE weight BETWEEN 100 AND 120;
```

This query shows patients whose weight is between 100 and 120, including both values.

Output :

first_name	last_name
Jiji	Sharma
Blair	Diaz
Thomas	ONeill
Sonny	Beckett
Tom	Halliwell
Jon	Doggett
Angel	Edwards
John	Farley
Temple	Russert
Don	Edwards
Gary	Ramotswe
Phil	Chester
Woody	Riviera
Rocky	Donahue
Shiro	Whale

Q5. Update the patients table for the allergies column. If the patient's allergies is null then replace it with 'NKA'

```
SELECT first_name, last_name,
CASE
WHEN allergies IS NULL THEN 'NKA'
ELSE allergies
END AS allergies
FROM patients;
```

This query updates the allergies column by replacing NULL values with 'NKA'. Since the server does not allow updates, a CASE statement can be used to display the updated result.

Output :

	first_name	last_name	allergies
▶	Donald	Waterfield	NKA
	Mickey	Baasha	Sulfa
	Jiji	Sharma	Penicillin
	Blair	Diaz	NKA
	Charles	Wolfe	Penicillin
	Sue	Falcon	Penicillin
	Thomas	ONeill	NKA
	Sonny	Beckett	NKA
	Sister	Spitzer	Penicillin
	Cedric	Coltrane	NKA
	Hank	Spencer	NKA

Q6. Show first name and last name concatenated into one column to show their full name.

```
SELECT CONCAT(first_name, ' ', last_name) AS
full_name
FROM patients;
```

This query joins the first name and last name together with a space in between and shows it as a single column called full_name.

Output :

full_name
Donald Waterfield
Mickey Baasha
Jiji Sharma
Blair Diaz
Charles Wolfe
Sue Falcon
Thomas ONeill
Sonny Beckett
Sister Spitzer
Cedric Coltrane
Hank Spencer
Sara di Marco

Q7. Show first name, last name, and the full province name of each patient.

```
SELECT p.first_name, p.last_name, pr.province_name
FROM patients p
JOIN province_names pr
ON p.province_id = pr.province_id;
```

This query joins the patients table with the province table to show the full province name for each patient.

Output :

first_name	last_name	province_name
Donald	Waterfield	Ontario
Mickey	Baasha	Ontario
Jiji	Sharma	Ontario
Blair	Diaz	Ontario
Charles	Wolfe	Ontario
Sue	Falcon	Ontario
Thomas	ONeill	Ontario
Sonny	Beckett	Nova Scotia
Sister	Spitzer	Ontario
Cedric	Coltrane	Ontario
Hank	Spencer	Ontario
Sara	di Marco	Ontario

Q8. Show how many patients have a birth_date with 2010 as the birth year.

```
SELECT COUNT(*)
FROM patients
WHERE YEAR(birth_date) = 2010;
```

This query counts how many patients were born in the year 2010.

Output :

COUNT(*)
55

Q9. Show the first_name, last_name, and height of the patient with the greatest height.

```
SELECT first_name, last_name, height
FROM patients
```

```
ORDER BY height DESC
LIMIT 1;
```

This query sorts patients by height in descending order and shows the top one, which is the tallest patient.

Output :

first_name	last_name	height
Sam	Haruko	226

Q10. Show all columns for patients who have one of the following patient_ids: 1, 45, 534, 879, 1000

```
SELECT *
FROM patients
WHERE patient_id IN (1, 45, 534, 879, 1000);
```

This query shows all details of patients whose IDs match any of the given values.

Output :

patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight
1	Donald	Waterfield	M	1963-02-12	Barrie	ON	NULL	156	65
45	Cross	Gordon	M	2009-03-20	Ancaster	ON	NULL	125	53
534	Don	Zatara	M	2008-01-11	Timmins	ON	NULL	136	67
879	Orla	Shawn	F	1967-09-24	Sarnia	ON	Penicillin	149	65
1000	Rick	Williams	M	1975-04-13	Hamilton	ON	Penicillin	176	127
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q11. Show the total number of admissions

```
SELECT COUNT(*) AS total_admissions
FROM admissions;
```

This query counts all the records in the admissions table to show the total number of admissions.

Output :

total_admissions
5067

Q12. Show all the columns from admissions where the patient was admitted and discharged on the same day.

```
SELECT *
FROM admissions
WHERE admission_date = discharge_date;
```

This query finds records where the admission date and discharge date are the same, meaning the patient was admitted and discharged on the same day.

Output :

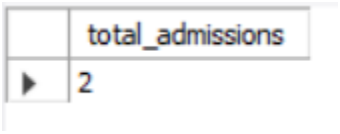
patient_id	admission_date	discharge_date	diagnosis	attending_doctor_id
1	2018-09-20	2018-09-20	Ineffective Breathin Pattern R/T Fluid Accumulatio	24
9	2018-12-31	2018-12-31	Ruptured Appendicitis	19
10	2019-02-27	2019-02-27	Lower Quadrant Pain	27
17	2019-03-04	2019-03-04	Diabetes Mellitus	9
28	2019-03-30	2019-03-30	Cancer Of The Stomach	26
31	2018-09-26	2018-09-26	Cardiovascular Disease	19
53	2018-10-24	2018-10-24	Urinary Tract Infection	8
54	2019-04-07	2019-04-07	Hypertension	21
70	2018-07-17	2018-07-17	Migraine	20
78	2018-06-17	2018-06-17	Hypertension	17
91	2018-08-30	2018-08-30	Congestive Heart Failure	3
92	2019-01-03	2019-01-03	Osteo Arthtitis Knee	5

Q13. Show the total number of admissions for patient_id 579.

```
SELECT COUNT(*) AS total_admissions
FROM admissions
WHERE patient_id = 579;
```

This query counts how many times the patient with ID 579 was admitted.

Output :



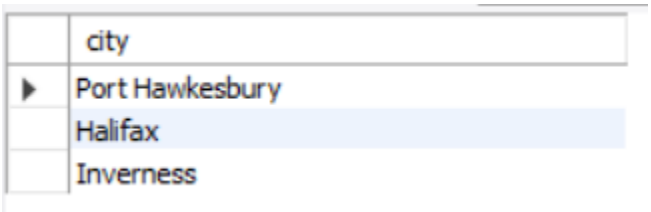
	total_admissions
▶	2

Q14. Based on the cities that our patients live in, show unique cities that are in province_id 'NS'?

```
SELECT DISTINCT city
FROM patients
WHERE province_id = 'NS';
```

This query shows all different cities where patients live in the NS province, removing duplicate city names.

Output :



	city
▶	Port Hawkesbury
	Halifax
	Inverness

Q15. Write a query to find the first_name, last name and birth date of patients who have height more than 160 and weight more than 70

```
SELECT first_name, last_name, birth_date
FROM patients
WHERE height > 160 AND weight > 70;
```

This query selects patients whose height is greater than 160 and weight is greater than 70, and shows their name and birth date.

Output :

first_name	last_name	birth_date
Mickey	Baasha	1981-05-28
Jiji	Sharma	1957-09-05
Blair	Diaz	1967-01-07
Thomas	ONeill	1993-01-31
Sonny	Beckett	1952-12-11
Sister	Spitzer	1966-10-15
Rick	Bennett	1977-01-27
Amy	Leela	1977-06-25
Tom	Halliwell	1987-08-01
Rachel	Winterbourne	1966-04-26
Jon	Doggett	1951-12-25
Angel	Edwards	1975-08-22

Q16. Show unique birth years from patients and order them by ascending.

```
SELECT DISTINCT YEAR(birth_date) AS birth_year
FROM patients
ORDER BY birth_year ASC;
```

This query gets all different birth years from the data and sorts them from smallest to largest.

Output :

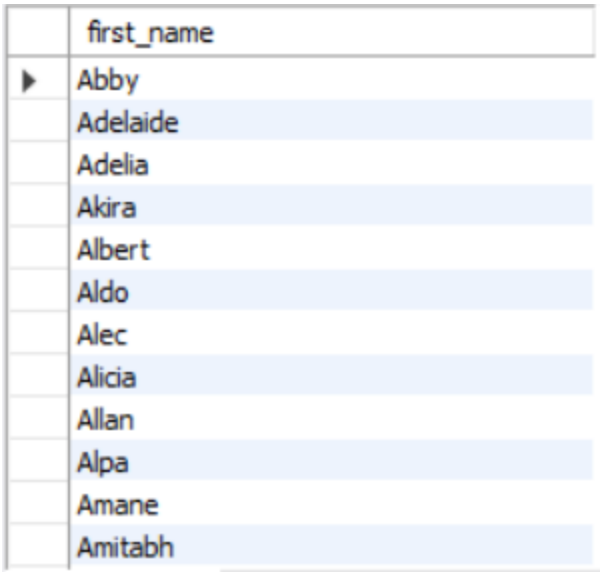
birth_year
1918
1923
1925
1926
1927
1928
1929
1931
1933
1934
1936
1937

Q17. Show unique first names from the patients table which only occurs once in the list.

```
SELECT first_name
FROM patients
GROUP BY first_name
HAVING COUNT(*) = 1;
```

This query groups the names and keeps only those that appear once in the table, removing duplicates.

Output :



first_name
Abby
Adelaide
Adelia
Akira
Albert
Aldo
Alec
Alicia
Allan
Alpa
Amane
Amitabh

Q18. Show patient_id and first_name from patients where their first_name start and ends with 's' and is at least 6 characters long.

```
SELECT patient_id, first_name
FROM patients
WHERE first_name LIKE 's%s'
AND LENGTH(first_name) >= 6;
```

This query checks names that start with ‘s’ and also end with ‘s’. It also makes sure the name has at least 6 characters. Then it shows the patient ID and first name.

Output :

patient_id	first_name
496	Spiros
629	Spiros
648	Stanislaus
1273	Stanislaus
1789	Seamus
1926	Stanislaus
1996	Stanislaus
2258	Spiros
2378	Stanislaus
2771	Stanislaus
3487	Stanislaus
NULL	NULL

Q19. Show patient_id, first_name, last_name from patients whos diagnosis is 'Dementia'. Primary diagnosis is stored in the admissions table.

```
SELECT p.patient_id, p.first_name, p.last_name
FROM patients p
JOIN admissions a ON p.patient_id = a.patient_id
WHERE a.diagnosis = 'Dementia';
```

Here we join patients with admissions to check diagnosis. Then we pick only those patients whose diagnosis is Dementia.

Output :

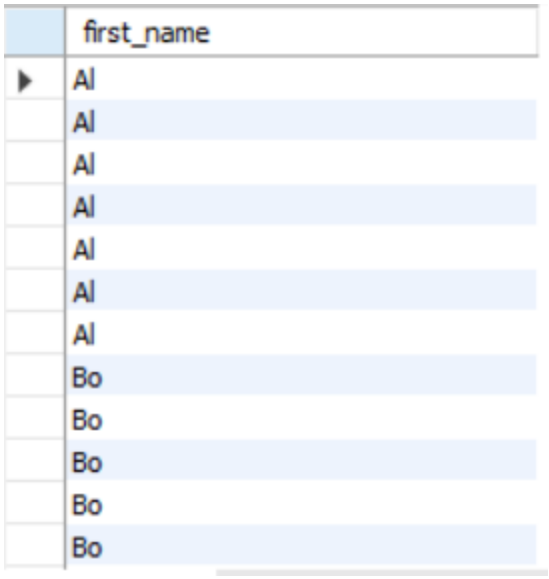
patient_id	first_name	last_name
160	Miranda	Delacour
178	David	Bustamonte
207	Matt	Celine
613	Jaki	Granger
836	Montana	Vimes
924	Simon	Spellman
1201	Irene	Murphy
1264	Jillian	Valentine
1402	Kathryn	Hallow
1491	Doris	McGrew
1585	Alex	Cantropus
1749	Alejandro	Mellie

Q20. Display every patient's first_name. Order the list by the length of each name and then by alphabetically.

```
SELECT first_name
FROM patients
ORDER BY LENGTH(first_name), first_name;
```

This shows all first names. First it sorts them by how long the name is, and if lengths are same, then it sorts alphabetically.

Output :



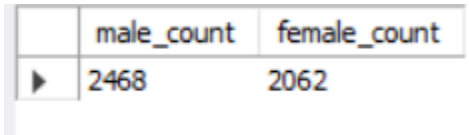
first_name
Al
Al
Al
Al
Al
Al
Al
Bo
Bo
Bo
Bo
Bo

Q21. Show the total amount of male patients and the total amount of female patients in the patients table. Display the two results in the same row.

```
SELECT
SUM(CASE WHEN gender = 'M' THEN 1 ELSE 0 END) AS
male_count,
SUM(CASE WHEN gender = 'F' THEN 1 ELSE 0 END) AS
female_count
FROM patients;
```

This query counts male and female patients separately and shows both counts in one row.

Output :



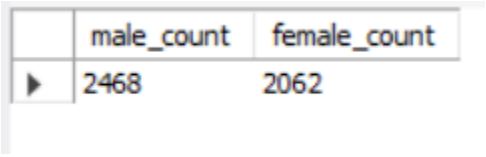
male_count	female_count
2468	2062

Q22. Show the total amount of male patients and the total amount of female patients in the patients table. Display the two results in the same row.

```
SELECT
SUM(CASE WHEN gender = 'M' THEN 1 ELSE 0 END) AS
male_count,
SUM(CASE WHEN gender = 'F' THEN 1 ELSE 0 END) AS
female_count
FROM patients;
```

This does the same as the previous one. It counts how many males and females are there and shows both in a single row.

Output :



male_count	female_count
2468	2062

Q23. Show patient_id, diagnosis from admissions. Find patients admitted multiple times for the same diagnosis.

```
SELECT patient_id, diagnosis
FROM admissions
GROUP BY patient_id, diagnosis
HAVING COUNT(*) > 1;
```

This groups records by patient and diagnosis, then shows only those where the same patient was admitted more than once for the same problem.

Output :

patient_id	diagnosis
137	Pregnancy
320	Pneumonia
1577	Congestive Heart Failure
2004	Left Shoulder Rotator Cuff Repair
2859	Severed Spine At C3
3012	Appendicitis
3367	Pyelonephritis
3468	Congestive Heart Failure
4083	Congestive Heart Failure
4121	Congestive Heart Failure
4363	Congestive Heart Failure

Q24. Show the city and the total number of patients in the city. Order from most to least patients and then by city name ascending.

```
SELECT city, COUNT(*) AS total_patients
FROM patients
GROUP BY city
ORDER BY total_patients DESC, city ASC;
```

This counts how many patients are in each city. Then it sorts the result by highest count first, and if counts are same, it sorts by city name.

Output :

city	total_patients
Hamilton	1938
Toronto	317
Burlington	276
Brantford	147
Ancaster	117
Stoney Creek	107
Cambridge	79
Dundas	79
Milton	65
Paris	58
Grimsby	55
Timmins	53

Q25. Show first name, last name and role of every person that is either patient or doctor. The roles are either "Patient" or "Doctor"

```
SELECT first_name, last_name, 'Patient' AS role
FROM patients
UNION
SELECT first_name, last_name, 'Doctor' AS role
FROM doctors;
```

This query combines data from patients and doctors into one list. It adds a role column to show whether the person is a patient or a doctor.

Output :

first_name	last_name	role
Donald	Waterfield	Patient
Mickey	Baasha	Patient
Jiji	Sharma	Patient
Blair	Diaz	Patient
Charles	Wolfe	Patient
Sue	Falcon	Patient
Thomas	ONeill	Patient
Sonny	Beckett	Patient
Sister	Spitzer	Patient
Cedric	Coltrane	Patient
Hank	Spencer	Patient
Sara	di Marco	Patient

Q26. Show all allergies ordered by popularity. Remove NULL values from query.

```
SELECT allergies, COUNT(*) AS total
FROM patients
WHERE allergies IS NOT NULL
GROUP BY allergies
ORDER BY total DESC;
```

This query counts how many times each allergy appears. It ignores NULL values and shows the most common allergies first.

Output :

	allergies	total
▶	Penicillin	1082
	Codeine	305
	Sulfa	157
	ASA	99
	Sulfa Drugs	71
	Peanuts	52
	Iodine	48
	Tylenol	42
	Bee Stings	40
	Valporic Acid	38
	Tetracycline	34
	Wheat	33

Q27. Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.

```
SELECT first_name, last_name, birth_date
FROM patients
WHERE YEAR(birth_date) BETWEEN 1970 AND 1979
ORDER BY birth_date ASC;
```

This query gets patients who were born between 1970 and 1979. Then it sorts them from the earliest birth date to the latest.

Output :

first_name	last_name	birth_date
Frances	Kobayakawa	1970-01-02
Sunny	Burrell	1970-01-07
Penelope	Beckett	1970-01-14
Deborah	Stewart	1970-01-14
Augusta	Decker	1970-01-22
Sookie	Brearly	1970-02-01
Temple	Wylie	1970-02-10
Deanna	Spano	1970-03-23
Jadu	Principal	1970-03-28
Betty	Stephens	1970-03-28
Jo	Sahid	1970-03-31
Santino	Hawker	1970-04-08
Andy	Guarnaccia	1970-04-10

Q28. We want to display each patient's full name in a single column. Their last_name in all upper letters must appear first, then first_name in all lower case letters. Separate the last_name and first_name with a comma. Order the list by the first_name in descending order. EX: SMITH,jane

```
SELECT CONCAT(UPPER(last_name), ',',
LOWER(first_name)) AS full_name
FROM patients
ORDER BY first_name DESC;
```

This query formats the name as LASTNAME,firstname. Last name is converted to uppercase and first name to lowercase. Then the list is sorted in descending order based on first name.

Output :

full_name
MILLER,zoe
CORBIE,ziva
KOBAYAKAWA,zenigata
OVERSTREET,zenigata
BENNETT,zen
MEPHESTO,zelda
MORRIS,zelda
THOMAS,zefram
FLUTE,zefram
MARONEY,zatanna
TYRELL,zatanna
CHE,zane
WONG,zane

Q29. Show the province_id(s), sum of height; where the total sum of its patient's height is greater than or equal to 7,000.

```
SELECT province_id, SUM(height) AS total_height
FROM patients
GROUP BY province_id
HAVING SUM(height) >= 7000;
```

This groups patients by province and adds up their heights. Then it shows only those provinces where the total height is 7000 or more.

Output :

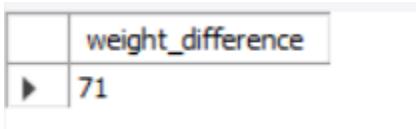
province_id	total_height
BC	7720
NS	9765
ON	678037

Q30. Show the difference between the largest weight and smallest weight for patients with the last name 'Maroni'

```
SELECT MAX(weight) - MIN(weight) AS
weight_difference
FROM patients
WHERE last_name = 'Maroni';
```

This query finds the highest and lowest weight for patients with last name Maroni and shows the difference between them.

Output :



	weight_difference
▶	71

Q31. Show all of the days of the month (1-31) and how many admission_dates occurred on that day. Sort by the day with most admissions to least admissions.

```
SELECT DAY(admission_date) AS day_of_month,
COUNT(*) AS total_admissions
FROM admissions
GROUP BY day_of_month
ORDER BY total_admissions DESC;
```

This query takes the day from each admission date and counts how many admissions happened on that day. Then it sorts the result from highest to lowest count.

Output :

day_of_month	total_admissions
11	184
4	184
9	183
2	180
12	179
6	179
16	177
21	174
13	173
28	173
7	167
8	166
18	166

Q32. Show all of the patients grouped into weight groups. Show the total amount of patients in each weight group. Order the list by the weight group descending. e.g. if they weight 100 to 109 they are placed in the 100 weight group, 110-119 = 110 weight group, etc.

```
SELECT FLOOR(weight/10)*10 AS weight_group,
COUNT(*) AS total_patients
FROM patients
GROUP BY weight_group
ORDER BY weight_group DESC;
```

This query puts patients into groups like 100–109, 110–119 by rounding down the weight. Then it counts how many patients are in each group and sorts the groups from highest to lowest.

Output :

weight_group	total_patients
140	6
130	59
120	191
110	426
100	507
90	403
80	478
70	633
60	685
50	443
40	202
30	126
20	165

Q33. Show patient_id, weight, height, isObese from the patients table. Display isObese as a boolean 0 or 1. Obese is defined as weight(kg)/(height(m)^2) >= 30. Weight is in units kg. Height is in units cm.

```
SELECT patient_id, weight, height,
CASE
WHEN weight / POWER(height/100, 2) >= 30 THEN 1
ELSE 0
END AS isObese
FROM patients;
```

This calculates BMI using weight and height. If the value is 30 or more, it marks the patient as obese (1), otherwise not obese (0).

Output :

patient_id	weight	height	isObese
1	65	156	0
2	76	185	0
3	106	194	0
4	104	191	0
5	10	47	1
6	5	43	0
7	117	180	1
8	105	174	1
9	95	173	1
10	61	157	0
11	74	158	0
12	46	145	0
13	77	146	1

Q34. Show patient_id, first_name, last_name, and attending doctor's specialty. Show only the patients who has a diagnosis as 'Epilepsy' and the doctor's first name is 'Lisa'.

```
SELECT p.patient_id, p.first_name, p.last_name,
d.specialty
FROM patients p
JOIN admissions a ON p.patient_id = a.patient_id
JOIN doctors d ON a.attending_doctor_id =
d.doctor_id
WHERE a.diagnosis = 'Epilepsy'
AND d.first_name = 'Lisa';
```

We join patients with admissions and doctors. Here we used attending_doctor_id because that is the actual column in the table. Then we filter patients who have Epilepsy and whose doctor's name is Lisa.

Output :

patient_id	first_name	last_name	specialty
468	Frank	Anderson	Obstetrician/Gynecologist
701	Precious	Ashton	Obstetrician/Gynecologist